

Measuring Before Claiming: Electrode Mapping, Spontaneous Burst Dynamics, and Plasticity Testing on Two Commercial Human Brain Organoid Cultures

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Abstract

Commercial cloud platforms now provide remote access to living human brain organoid cultures, opening experimental neuroscience to researchers without laboratory affiliation. What such platforms do not provide is a characterization of the specific tissue a given user receives. This report documents a 12-day measurement campaign on two human neurosphere cultures accessed through the FinalSpark Neuroplatform, in which tissue characterization was treated as the primary object of study, ahead of any experimental protocol.

Four properties were established. First, functional electrode coverage was sparse and culture-specific: of 32 available electrodes, 7 carried measurable activity in the first culture and 10 in the second, with partial but incomplete overlap between them. Second, both cultures produced spontaneous synchronized bursts with no stimulation applied; in the second culture, 39,031 consecutive readings across 5.5 hours yielded 404 discrete events at matched sampling rate, approximately twice the rate measured in the first culture under identical protocol. Third, event magnitudes were incompatible with a power-law distribution across five independent detection thresholds, tested against both exponential and lognormal alternatives, indicating that this tissue did not operate in the near-critical regime described for cortical preparations. Fourth, event timing was strongly clustered rather than Poisson-distributed, with a coefficient of variation between 2.35 and 2.67 depending on sampling resolution.

Spike-timing-dependent plasticity was tested using an isolated single-pair protocol with millisecond-scale pairing intervals. Small-sample measurements produced a directionally symmetric effect consistent with the classical STDP curve. When the same protocol was repeated with larger samples and a drift-corrected measurement ratio, the effect did not reach statistical significance in either culture across three electrode pairs. The measurement pipeline developed to reach that conclusion (burst detection and rejection, ratio-based drift correction, and sampling-rate matching between cultures) is reported in full, along with two analysis errors identified and corrected during the work.

All raw data, analysis code, and an independent verification script are released openly.

Keywords: brain organoid, microelectrode array, spontaneous activity, neuronal avalanche, spike-timing-dependent plasticity, measurement methodology, organoid intelligence

1 Introduction

1.1 Cloud-accessible organoid platforms

Human brain organoids grown from induced pluripotent stem cells produce spontaneous electrical activity and can be interfaced with microelectrode arrays (MEAs). Over the past several years, this capability has moved from specialist laboratories to commercial platforms that offer remote, scheduled access to live cultures. A researcher can now write code on a laptop, connect to a cloud environment, and stimulate living neural tissue in another country within the same session.

This change removes a barrier that has historically been decisive. Organoid electrophysiology previously required a laboratory, a culture facility, and the personnel to maintain both. Remote platform access replaces these with a subscription. The present work was conducted entirely through such an arrangement, by a single researcher with no institutional affiliation, no laboratory access, and no co-investigators.

1.2 What platform access does not provide

Remote access delivers a connection and an API. It does not deliver a characterization of the specific tissue at the other end. The user receives a culture identifier, a set of electrode channels, and a booking window. Which electrodes carry signal, how the tissue behaves when left alone, and how stable its baseline is across hours or days are properties the user must determine independently.

These properties are not incidental. Every closed-loop protocol assumes that stimulation reaches responsive tissue, that measured changes reflect the intervention and not the tissue's own dynamics, and that the baseline against which effects are measured holds still long enough for the comparison to mean something. Each of these assumptions was tested in the present work, and each required correction.

1.3 Scope and framing

This report treats tissue characterization as the primary result, ahead of the protocols it was meant to support. The measurements described here were conducted over 12 days on two cultures, with a total of approximately 40 hours of platform access. The data comprise electrode-level activity maps, a 5.5-hour continuous passive recording spanning two sessions, and a series of plasticity tests conducted under progressively stricter statistical criteria.

The intent is to provide a concrete, numerically specified account of what one user measured on one platform, including the measurement infrastructure required to reach reliable numbers. Two cultures from one supplier do not establish general properties of organoid tissue. They do establish what a researcher should expect to find, and what they should check, before interpreting any result obtained on such a platform.

2 Methods

2.1 Platform and hardware interface

Experiments were conducted on the FinalSpark Neuroplatform, a cloud-accessible system providing scheduled sessions with living human neurosphere cultures maintained on multi-electrode arrays. Access was arranged under a shared-access subscription, which allocates dedicated time slots to individual users while the same cultures are used by other researchers between sessions.

Two cultures were used across the campaign, designated fs644 and fs652 by the platform. Culture transitions were performed by platform staff for maintenance reasons and were not under user control. The transition from fs644 to fs652 occurred on 11 September 2026 and was announced through an automated platform notification.

Hardware control used the platform's `neuroplatformv2` Python package, available only within the provider's cloud notebook environment. Stimulation was delivered through `IntanController._send_stimparam` followed by a trigger dispatch; spike counts were read through `IntanController._count_spike` over a specified time window.

2.2 API variant detection

The `neuroplatformv2` interface was found to differ between workspace configurations. In one configuration, `TriggerController` required an account email argument and spike reading used `_count_spike(ms)`; in another, `TriggerController` took no arguments and reading required a paired `_set_count/_read_count` sequence.

Rather than hardcoding either variant, the hardware adapter performs runtime introspection using `inspect.signature` and `dir`, selecting the appropriate call pattern at connection time:

```
def detect_api():
    from neuroplatformv2.core.intan import IntanController
    from neuroplatformv2.core.trigger import TriggerController
    info = {}
    trig_sig = inspect.signature(TriggerController.__init__)
    info["trigger_needs_email"] = (
        "email" in trig_sig.parameters
        and trig_sig.parameters["email"].default is inspect.Parameter.empty
    )
    intan_methods = [m for m in dir(IntanController)
                     if not m.startswith("__")]
    if "_count_spike" in intan_methods:
        info["read_method"] = "count_spike"
    elif "_set_count" in intan_methods and "_read_count" in intan_methods:
        info["read_method"] = "set_read_count"
    return info
```

This allows the same experimental code to run unmodified across workspace configurations.

2.3 Stimulation parameters

Stimulation used biphasic pulses with negative-first polarity, 100 μ s per phase, 1.0 μ A amplitude. Multi-electrode patterns were delivered by assigning per-electrode `trigger_delay` values in microseconds, allowing sub-millisecond control of relative timing within a single trigger event.

A 10-second wait after `_send_stimparam` and before trigger dispatch was required by the hardware and confirmed with platform technical staff. This constraint sets the practical floor on trial duration: a closed-loop trial that includes a stimulation phase and a reward-delivery phase carries at least three such waits, producing a measured mean trial duration of 39.5 seconds. Protocols were sized against this constraint, not against nominal trial counts.

2.4 Passive recording protocol

Passive recordings applied no stimulation. Each reading called `_raw_read(duration_s=0.5)`, returning per-electrode spike counts over a 0.5-second window. Readings were taken back-to-back, producing a median inter-reading interval of 0.503 s (verified from timestamps).

The activity measure used throughout is the summed spike count across the culture's active electrode set within a single reading window, referred to as the active-zone total.

Long recordings were made resilient to session boundaries by writing every reading to disk with a Unix timestamp and reloading the accumulated file at the start of each subsequent session. This allowed a single continuous dataset to span two bookings separated by a 12.21-hour gap.

2.5 Event detection

Events were defined as threshold crossings in the active-zone total, with consecutive crossings within 3.0 seconds merged into a single event. This merge rule matters: without it, a single burst sampled at high rate produces multiple counted events, and event rates become a function of sampling frequency, not of tissue behaviour.

For cross-culture comparison, event rates were computed at matched sampling resolution by downsampling the higher-rate recording, since event detection is sensitive to sampling rate (Section 3.5).

2.6 Burst rejection during measurement

Once spontaneous bursts were characterized (Section 3.2), a rejection filter was integrated directly into the hardware adapter's read path, and not applied as a post-hoc analysis step:

```
def _read_spikes(self, duration_s):
    for attempt in range(BURST_MAX_RETRIES):
        counts = self._raw_read(duration_s)
        active_total = sum(counts[e] for e in ACTIVE_ELECTRODES)
        if active_total <= BURST_THRESHOLD:
            break
        time.sleep(BURST_RETRY_WAIT_S)
    return counts
```

Readings exceeding the threshold are discarded and retaken after a 3-second delay, chosen to exceed the measured mean burst duration. Because the filter sits in the read path, every downstream protocol inherits it without modification.

2.7 Drift-corrected plasticity measurement

Baseline activity was found to change over hours (Section 3.3), which affects any pre/post comparison: a post-training measurement taken later than its pre-training reference will differ from it even with no intervention effect.

To separate intervention effects from baseline movement, each plasticity measurement paired a probe with an immediately adjacent unstimulated baseline reading, and the analysis used their ratio:

```
def measure_point():
    a = probe_response(driver_electrode)  # driver stimulated, readout measured
```

```
b = read_baseline()          # no stimulation, active-zone total
return a / (b + 0.5)
```

The additive constant prevents division instability when the baseline reading is near zero. Under this measure, a change in overall tissue excitability moves numerator and denominator together and leaves the ratio unchanged, while a pathway-specific change moves the ratio.

2.8 STDP pairing protocol

Plasticity induction used a single driver electrode and a single readout electrode, with the driver fired 20 ms before the readout (pre-before-post, the potentiating direction of the classical STDP curve). The reverse direction fired the readout first.

Measurement consisted of a pre-training block, a training block of paired stimulations, and a post-training block. Probes during pre- and post-training blocks stimulated only the driver electrode and measured only the readout electrode, with no pairing applied.

2.9 Statistics

Group comparisons used Welch's t-statistic with a normal-approximation 95 % confidence interval on the difference of means. An effect was treated as established only when the interval excluded zero.

Power-law fitting used maximum-likelihood estimation following Clauset, Shalizi and Newman (2009), implemented in the `powerlaw` package (Alstott, Bullmore and Plenz, 2014). Model comparison used log-likelihood ratios against exponential and lognormal alternatives, with the sign of the ratio indicating which distribution fits better and the associated p -value indicating whether that difference is distinguishable from noise.

2.10 Data and code availability

All raw data, analysis code, and an independent verification script that recomputes every reported number from the raw file are released (Section 8).

3 Results

3.1 Functional electrode coverage is sparse and culture-specific

Both cultures were interfaced through 32-electrode arrays. In fs644, repeated stimulation and recording across 16 trials identified 7 electrodes carrying measurable activity: channels 8, 9, 10, 11, 12, 14 and 15. The remaining 25 channels returned zero or near-zero counts regardless of stimulation pattern.

When the platform transitioned to fs652, the same mapping procedure was repeated before any experimental protocol was run. Passive readings across all 32 channels gave the following per-electrode means over three consecutive readings:

Electrode	Mean count	Electrode	Mean count
8	0.67	15	3.00
9	1.00	17	0.33
10	0.67	19	4.33
12	1.00	0	0.33
14	2.67	3	0.33

Ten channels carried measurable activity in fs652 against seven in fs644. Six channels were common to both maps (8, 9, 10, 12, 14, 15). Channel 11, active in fs644, was silent in fs652; channels 0, 3, 17 and 19 were active in fs652 but not in fs644.

Two conclusions follow. Functional coverage is a fraction of nominal coverage, which bears directly on how stimulation patterns should be designed. And the map is a property of the individual culture, not of the platform, so it requires re-derivation whenever the tissue changes.

3.2 Both cultures produce spontaneous synchronized bursts

A one-hour passive recording on fs644 with no stimulation applied yielded 3,600 readings. The active-zone total had a mean of 12.9 with a standard deviation of 18.8, a dispersion larger than the mean, driven by a small number of extreme values. The maximum active-zone reading was 640, and the maximum across all 32 channels reached 3,053.

Grouping consecutive threshold crossings into discrete events (Section 2.5) gave 37 events in one hour. Inter-event intervals averaged 92.2 s with a range of 7 to 382 s. Mean event duration was approximately 1.4 s.

The magnitude of these events relative to baseline is the operationally relevant figure: a burst reaching 640 against a baseline mean of 12.9 exceeds it by a factor of roughly 50. Any measurement window of a few seconds that happens to contain a burst will be dominated by it.

This finding reframed a pattern that had appeared across earlier experimental sessions, in which effects appeared in small samples and reversed direction on repetition. A burst landing by chance in one condition and not another produces exactly that signature. Once detected and filtered, measurement variability dropped substantially: in one calibration, per-condition standard deviations fell from the 4–9 range to 0.8–2.9.

3.3 Baseline activity shifts across sessions

The continuous recording spanning two sessions provides a direct measurement of baseline stability. Session one (11 September, 19:02–20:47) contributed 12,445 readings; session two (12 September, 09:00–12:45) contributed 26,586. The sessions were separated by 12.21 hours during which no user access occurred.

Measure	Session 1	Session 2
Readings	12,445	26,586
Mean active-zone total	6.74	16.22
Median	6.0	14.0
Within-session trend (r)	−0.044	+0.038

The mean more than doubled between sessions. The shape of that change is informative: within-session trends were negligible in both sessions, while the transition between the last 500 readings of session one (mean 6.90) and the first 500 of session two (mean 14.88) shows a step of +7.98. A Mann-Whitney U test on the full session distributions returns $p \approx 0$.

The change therefore takes the form of a step between sessions, with no matching gradual drift within them. Under the shared-access model, cultures are used by other researchers between bookings, which the platform confirmed in writing; this provides a plausible mechanism, though the present data cannot attribute the step to a specific cause.

A separate and much smaller within-session drift is also present. Averaging session two into 29 consecutive 15-minute windows and fitting a line gives $r = 0.797$ across those windows. The underlying effect is small: the total change across 7.25 hours is 1.39 units, approximately 13 % of the session standard deviation of 10.53, and the correlation computed on raw unaveraged readings is $r = +0.038$. Window averaging suppresses noise and makes a weak trend appear visually and statistically stronger than its effect size warrants; both figures are reported here for that reason.

3.4 Event magnitudes are incompatible with power-law scaling

The continuous recording (39,031 readings, 5.5 hours of active recording time confirmed from timestamps) was analyzed for avalanche statistics following the framework of Beggs and Plenz (2003), which characterizes near-critical cortical dynamics by power-law-distributed event magnitudes.

Event extraction required a methodological adjustment. The standard definition bounds avalanches by intervals of zero activity. In this tissue, only 15 of 39,031 readings were exactly zero (0.04 %), so zero-bounded extraction grouped multi-hour stretches into single events, the largest spanning 27,373 consecutive readings. That extraction is not physically meaningful and its output was discarded.

Threshold-based extraction was used instead, defining events as contiguous runs above a threshold with magnitude measured as the summed excess above it. Model comparison was run at five thresholds spanning the 75th to well above the 99th percentile of the activity distribution:

Threshold	Events	LLR vs exponential	LLR vs lognormal
10	5,919	−34.62	−34.08
16	5,426	−132.50	−170.90
22	2,773	−39.26	−36.01
29	1,556	−40.06	−35.93
40	851	−43.66	−38.01

All ratios are negative with $p < 0.0001$, indicating that both exponential and lognormal distributions describe the magnitude data better than a power law at every threshold tested.

This is a clean negative result on a specific, pre-registered question. In this culture, over 5.5 hours of continuous recording, spontaneous event magnitudes do not exhibit the scaling signature associated with near-critical dynamics. The consistency across five thresholds and two alternative distributions makes the conclusion robust to the arbitrary choices involved in event definition.

3.5 Event timing is clustered rather than random

While magnitudes are compatible with random generative processes, timing is not. Inter-event intervals were computed at three sampling resolutions:

Sampling	Events detected	Mean interval (s)	CV
Full (2/s)	727	26.9	2.66
Halved (1/s)	404	48.2	2.35
Quartered (0.5/s)	214	91.0	2.67

A Poisson process yields a coefficient of variation of 1.0. Values between 2.35 and 2.67 indicate substantial clustering: events arrive in groups separated by longer quiet periods. The result holds across a fourfold range of sampling resolution.

The combination is worth stating directly. Event magnitudes are consistent with an unstructured random process, while event timing carries clear structure. These are separable properties of the same spontaneous activity, and a characterization that reports only one of them is incomplete.

Sampling resolution has a strong effect on measured event rate: 132.2, 73.5 and 38.9 events per hour at the three resolutions above. Any cross-study or cross-culture comparison of burst rates that does not match sampling resolution is comparing measurement protocols as much as tissue.

3.6 Cross-culture burst rate comparison

With sampling resolution matched to the fs644 protocol (1 reading/s) and the same 3-second event-merging rule applied to both, fs652 produced 404 events across 5.5 hours, or 73.5 events per hour, against 37.0 events per hour in fs644. Mean event duration was 0.55 s in fs652 against approximately 1.4 s in fs644.

The second culture therefore produced bursts roughly twice as often, each lasting roughly half as long. Both cultures were human neurospheres from the same supplier, measured through the same API with the same code and the same threshold.

Two cultures do not establish a distribution, and this comparison is offered as a measured instance and not a general property. Its value lies in the protocol matching: because both numbers were produced by identical event definitions at identical sampling resolution, the difference between them is a property of the tissue, not of the measurement.

3.7 Isolated-pair plasticity testing

Spike-timing-dependent plasticity was tested using the isolated single-pair protocol (Section 2.8), which fires one driver electrode 20 ms before one readout electrode and measures whether the readout's response to the driver alone changes.

Initial measurements on fs644 using electrode pair 9→14 produced a directionally symmetric result. Forward pairing (driver first) gave a combined difference of +0.83 across 23 measurements; reverse pairing (readout first) gave −0.80 across 15 measurements. The symmetry and the sign of each direction match the classical STDP curve, in which pre-before-post pairing potentiates and post-before-pre pairing depresses.

The protocol was then repeated with larger samples and the drift-corrected ratio measure. Three tests

were run:

Culture	Pair	n (pre/train/post)	Ratio change	95 % CI
fs644	9→14	20 / 40 / 20	+0.083	[−0.033, +0.198]
fs652	9→14	40 / 60 / 40	−0.098	[−0.244, +0.048]
fs652	12→19	8 / 15 / 8	+0.106	[−0.273, +0.486]

All three confidence intervals contain zero. Interim measurements taken during the 40-pairing training block of the first test showed no upward trend across training (ratios of 0.57, 0.67, 0.40, 0.46, 0.40, 0.62, 0.32, 0.42 at successive checkpoints).

The larger-sample tests establish an upper bound on any effect present under these conditions. The measurements were made with burst rejection active and drift correction applied, on two independent cultures, across three electrode pairs, with sample sizes two to five times those of the initial measurements. Under those conditions, pairing-induced change in the driver-to-readout ratio was not distinguishable from zero.

The initial small-sample result and the larger-sample tests are both reported here because the relationship between them is itself informative. An effect that appears with directional symmetry at $n = 15$ –23 and does not survive at $n = 40$ –60 with drift correction is a well-documented pattern in fields where effect sizes are small relative to measurement variance. The present data provide a worked instance of it in organoid electrophysiology, with the full measurement chain available for inspection.

3.8 Testing a network convergence hypothesis

Mapping of fs652 identified electrode 19 as carrying the highest passive activity of any channel. A systematic stimulation survey then found that electrode 19 gave the strongest secondary response to every driver tested, within a narrow range:

Driver	Response at 19	Driver	Response at 19
8	5.3	16	5.8
9	5.4	17	5.2
10	4.9	18	5.0
12	5.2	20	5.0
14	5.5	21	5.4
15	4.9		

Across 11 drivers, responses at electrode 19 spanned 4.9 to 5.8 with a standard deviation of 0.27. A convergence interpretation would treat this as evidence that electrode 19 sits at a functional confluence of the network.

A control measurement settled the question. Passive recording of electrode 19 with no stimulation applied gave a mean of 5.27 across 15 readings (range 3–7), statistically indistinguishable from its mean response of 5.20 under stimulation from any driver.

For comparison, the same survey read out at electrode 14 gave driver-dependent responses ranging from 1.80 to 4.80 with a standard deviation of 1.05, showing the driver-specific variation that a genuinely stimulus-driven readout produces.

Electrode 19 therefore fires at a consistent rate that does not depend on external stimulation. The uniformity across drivers reflects an autonomous, pacemaker-like firing rate, with no evidence of network

convergence. The hypothesis was proposed, tested against a specific alternative, and resolved in favour of the alternative within a single session.

This result carries a practical implication for readout selection on any MEA-based protocol: an electrode chosen for high activity may be measuring its own baseline and not the response to stimulation, and a passive control is sufficient to distinguish the two.

3.9 Two analysis errors identified and corrected

Two errors in the author’s own analysis were found during a systematic re-derivation of all reported values from the raw data, and both are documented here because each represents a general and recurring analytical hazard.

Unmatched protocols in cross-culture comparison. The initial fs644-versus-fs652 burst rate comparison reported 154.7 events per hour against 37.0, a ratio of approximately 4.2. Re-derivation revealed that the 3-second event-merging rule applied to fs644 had not been applied to fs652, and that the two recordings were made at different sampling rates (1/s and 2/s). Applying identical event definitions at matched sampling resolution gives 73.5 events per hour, a ratio of approximately 2.0. The direction of the difference is unchanged; its magnitude was overstated by a factor of two.

Window averaging inflating a weak trend. The within-session drift in session two was initially reported as a linear trend with $r = 0.797$ and $r^2 = 0.636$. Those figures were computed on 29 window means and not on raw readings. The raw correlation is $r = +0.038$, and the total change across 7.25 hours is 1.39 units against a session standard deviation of 10.53. The trend is present and the direction is reliable; describing it as strong was not supported by the effect size.

Both errors were found by recomputing every reported number from the raw file with independently written code, checking stated assumptions against the data (sampling interval and total recording duration were verified from timestamps and both proved correct), and testing each finding for robustness against the arbitrary choices in its derivation. Three of the five quantitative findings passed unchanged; one required a magnitude correction and one required a reframing of its strength.

4 Measurement infrastructure

The findings above required a measurement pipeline that did not exist at the start of the campaign. Its components are described here because they are reusable and because each was added in response to a specific measured problem, without being anticipated in advance.

Runtime API detection (Section 2.2) removed a class of connection failures arising from workspace-dependent interface differences.

Active-zone restriction. Stimulation patterns were initially distributed across electrode indices starting from zero, which directed a substantial fraction of stimulation to channels carrying no measurable activity. Encoding was changed to draw only from the measured active set. The change is parameterized so that simulation behaviour is unaffected when no active set is supplied.

Burst rejection in the read path (Section 2.6) ensures that every protocol built on the hardware adapter inherits burst filtering without modification.

Ratio-based drift correction (Section 2.7) separates pathway-specific change from tissue-wide excitability change.

Interleaved measurement ordering. Multi-condition measurements were reordered from blocked (all repetitions of condition A, then all of B) to round-robin (one repetition of each condition per cycle). Under blocked ordering, a baseline that moves across the measurement session confounds condition with time; round-robin distributes any drift equally across conditions.

Sampling-rate matching for any cross-recording comparison, following the demonstration in Section 3.5 that event rate depends strongly on sampling resolution.

File-backed session continuity. Long recordings write each reading to disk with a Unix timestamp and reload the accumulated file at session start, allowing a single dataset to span multiple bookings. This produced the 39,031-reading dataset across a 12.21-hour gap.

5 Discussion

5.1 Characterization as a result

The measurements reported here were originally intended as preparation for closed-loop learning experiments. They became the primary result because each characterization step changed what the subsequent experiments could validly claim.

Sparse electrode coverage determines where stimulation can be delivered. Spontaneous bursting at 50 times baseline determines what measurement windows are safe. Between-session baseline steps of more than twofold determine whether comparisons across sessions are interpretable. None of these properties was available from the platform, and each was recoverable with a few hours of passive recording.

For researchers beginning work on commercial organoid platforms, the practical recommendation is direct: measure the electrode map, characterize spontaneous activity with stimulation off, and quantify baseline stability across the intended experimental timescale, before running any protocol whose interpretation depends on those properties.

5.2 Interpretation of the plasticity results

The larger-sample plasticity tests place an upper bound on effects obtainable under the conditions tested: single-pair pairing at 20 ms intervals, 15–60 pairings, measured through the driver-to-readout ratio with burst rejection active.

Several conditions were not explored and remain open. Pairing intervals other than 20 ms were not tested, and the STDP time window is narrow enough that the choice matters. Pairing counts above 60 were not reached within session limits. The readout measure, a spike count within a 0.5 s window at a single electrode, is coarse relative to measures available with full waveform access.

The result is therefore specific, and bounded by those conditions: under this protocol, on these two cultures, with these sample sizes, pairing-induced change was not distinguishable from zero. It does not establish that the tissue lacks plasticity.

5.3 Criticality

The avalanche analysis provides a clear negative result on a well-specified question. Across five thresholds, event magnitudes were better described by exponential and lognormal distributions than by a power law, with likelihood ratios ranging from -34 to -171 and $p < 0.0001$ throughout.

The clustering result (CV 2.35–2.67) is the complementary finding. Magnitudes carry no scaling structure while timing carries clear temporal structure, and the two properties are separable in the same dataset. A characterization reporting only magnitude statistics would describe this tissue as generating random events, which the timing statistics contradict.

The zero-bounded extraction problem (Section 3.4) is worth noting for others analyzing similar data. Where a preparation rarely reaches exact silence, the standard avalanche definition produces events spanning hours, and any downstream distributional test on those events is meaningless regardless of the p -value it returns.

5.4 The value of the error record

Section 3.9 documents two errors in the author’s own analysis. Both are instances of general hazards: comparing measurements made under different protocols, and reporting a correlation computed on smoothed data as though it characterized the underlying signal.

Reporting them serves two purposes. It allows any reader to distinguish the figures that survived re-derivation from those that required correction. And it demonstrates that the verification procedure (re-computing every number from raw data with independent code, checking stated assumptions against timestamps, and testing robustness against arbitrary analytical choices) detects real errors when applied, including errors that favoured the author’s initial interpretation.

6 Limitations

Two cultures from a single supplier, accessed through a single platform, do not support general claims about brain organoid tissue. Every quantitative result here is a measured instance.

Culture selection was not under user control. Under shared access, the same cultures are used by other researchers between sessions, and the platform confirmed that experiments are regularly performed on shared cultures between bookings. Between-session changes in baseline (Section 3.3) cannot be attributed to a specific cause with the present data.

The readout measure is a spike count within a 0.5 s window summed across an electrode set. Waveform-level features, spike sorting, and sub-window timing were not available through the interface used.

Total access was approximately 40 hours across 12 days, distributed across sessions of 2–4 hours. Protocols requiring longer continuous observation, or measurement of retention over days, were not feasible within that budget.

The plasticity tests explored a single pairing interval (20 ms) and pairing counts up to 60. The parameter space of STDP induction is substantially larger.

7 Conclusion

Four properties of two commercial human brain organoid cultures were established through direct measurement. Functional electrode coverage was sparse (7 of 32 channels in one culture, 10 of 32 in the other) and culture-specific, with partial overlap. Both cultures generated spontaneous synchronized bursts reaching tens of times baseline, at rates of 37.0 and 73.5 events per hour under matched protocol. Event magnitudes were incompatible with power-law scaling across five thresholds and two alternative distributions, while event timing was strongly clustered (CV 2.35–2.67). Baseline activity changed by more than

a factor of two across a 12-hour interval between sessions, in the form of a step and not a within-session drift.

Spike-timing-dependent plasticity was tested with an isolated single-pair protocol under progressively stricter conditions. Small-sample measurements produced a directionally symmetric effect matching the classical STDP curve; larger-sample tests with drift correction, across two cultures and three electrode pairs, returned confidence intervals containing zero.

The measurement infrastructure developed to reach these conclusions (runtime API detection, active-zone restriction, burst rejection integrated into the read path, ratio-based drift correction, interleaved measurement ordering, sampling-rate matching, and file-backed session continuity) is released with the data. So is the record of two analysis errors found by systematic re-derivation and corrected.

The work was conducted by one researcher, without laboratory access or institutional affiliation, through a commercial cloud platform. That it was possible at all is a consequence of how organoid access has changed. What the measurements show is that platform access supplies a connection, and everything downstream of that connection is the user's responsibility to establish.

8 Data and code availability

Raw data. The complete 39,031-reading continuous recording (`burst_izleme_veri.json`), electrode mapping measurements, and all plasticity test results in JSON format.

Analysis code. All measurement scripts, the hardware adapter with burst rejection, and the analysis pipeline.

Verification script. `dogrulama_v2.py` recomputes every quantitative result reported in this paper directly from the raw data file and prints each alongside its expected value. Running it requires Python 3, NumPy, and the `powerlaw` package.

Repository: <https://github.com/Metin1558>

Project site: <https://metin.science>

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Platform access was provided by FinalSpark under a shared-access subscription. The author thanks the FinalSpark team for technical support during live sessions, including confirmation of hardware timing constraints and identification of a facility event that explained one baseline transition.

Use of AI assistance. A large language model (Anthropic Claude) was used during this project for code drafting, analysis scripting, and manuscript preparation. All experimental design decisions, all hardware sessions, and all interpretation of results were made by the author. Every quantitative value reported here was recomputed from the raw data by an independent verification script (Section 8), and the two analysis errors documented in Section 3.9 were found through that verification process. The author takes full responsibility for the content.

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